

Roxelle Sales & Operations Analytics Project Summary

Executive Summary

This project was developed for Roxelle Nigeria Ltd. as part of a Business Intelligence and Data Analytics capstone exercise using Microsoft Power BI. The objective of the project was to transform raw retail transaction data into meaningful business insights capable of supporting executive decision-making across sales, customer management, inventory monitoring, and promotional performance.

The solution focused on analysing retail operations across multiple branches using interactive dashboards, DAX calculations, and data modelling techniques. The final report provides management with visibility into revenue performance, profitability, customer purchasing behaviour, product trends, and discount effectiveness.

Project Objectives

The major objectives of the project were to:

- ✓ Analyse overall sales and profit performance
- ✓ Identify high-performing and low-performing branches
- ✓ Understand customer purchasing behaviour and loyalty trends
- ✓ Monitor product performance and inventory health
- ✓ Evaluate the effectiveness of discounts and promotions
- ✓ Build interactive dashboards for executive reporting
- ✓ Generate actionable business recommendations

Tools & Technologies Used

The project was implemented using:

- Microsoft Power BI Desktop
- Power Query Editor
- DAX (Data Analysis Expressions)
- Power BI Service
- GitHub

Dataset Overview

The project utilised five structured CSV datasets:

Dataset	Description
branches.csv	Branch information
customers.csv	Customer demographic data
products.csv	Product and inventory details

Dataset	Description
sales_transactions.csv	Transactional sales records
date.csv	Calendar/date dimension

The datasets were cleaned and transformed using Power Query before being modelled into a star schema structure.

Data Modelling

A star schema model was implemented to improve dashboard performance and analytical flexibility.

Fact Table

- ✓ sales_transactions

Dimension Tables

- ✓ customers
- ✓ products
- ✓ branches
- ✓ date

Relationships were configured using one-to-many cardinality with single-direction filtering.

Dashboard Pages Developed

1. Sales Performance Dashboard

This dashboard focused on:

- ✓ Total revenue
- ✓ Total profit
- ✓ Profit margin
- ✓ Branch revenue analysis
- ✓ Revenue trends over time
- ✓ Category contribution analysis

2. Customer Behaviour Analysis

This dashboard analysed:

- ✓ Customer segmentation
- ✓ Basket size
- ✓ Loyalty card performance
- ✓ Age and gender distribution
- ✓ Top customers by revenue

3. Product & Inventory Analysis

This dashboard monitored:

- Top and bottom performing products
- Product return rates

- Inventory reorder levels
- Product category contribution

4. Discount & Promotions Audit

This dashboard evaluated:

- ✓ Revenue lost through discounting
- ✓ Discount impact by payment method
- ✓ Relationship between discounts and return rates
- ✓ Branch discount performance

Key Business Insights

The analysis revealed several important insights:

- ✓ Certain branches generated high revenue but experienced reduced profitability due to excessive discounting.
- ✓ Loyalty card customers showed stronger purchasing patterns compared to non-loyalty customers.
- ✓ Some products consistently recorded high return rates and may require quality review.
- ✓ Food & Beverages contributed the largest share of overall revenue.
- ✓ Some products were approaching reorder-level thresholds, indicating potential stock risks.
- ✓ High discounts did not always translate into increased profitability.

Recommendations

Based on the analysis, the following recommendations were made:

- ✓ Reduce unnecessary discounting in low-profit branches.
- ✓ Improve inventory monitoring for products nearing reorder levels.
- ✓ Strengthen loyalty card campaigns to increase customer retention.
- ✓ Review products with consistently high return rates.
- ✓ Focus promotional spending on high-performing product categories.
- ✓ Implement regular executive dashboard monitoring for decision-making.

Conclusion

The project successfully transformed transactional retail data into a professional business intelligence solution using Microsoft Power BI.

The final solution demonstrates:

- ✓ Data cleaning and transformation
- ✓ Data modelling
- ✓ DAX calculations
- ✓ Interactive dashboard development
- ✓ Executive insight generation

This project highlights the importance of business intelligence tools in improving operational visibility and supporting data-driven decision-making within retail organisations.